Kernix 0.11.1 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

46 workloads

149 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

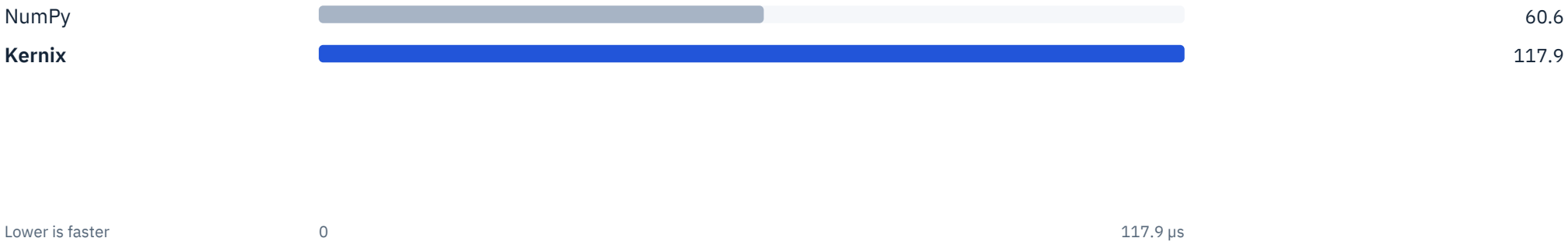
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



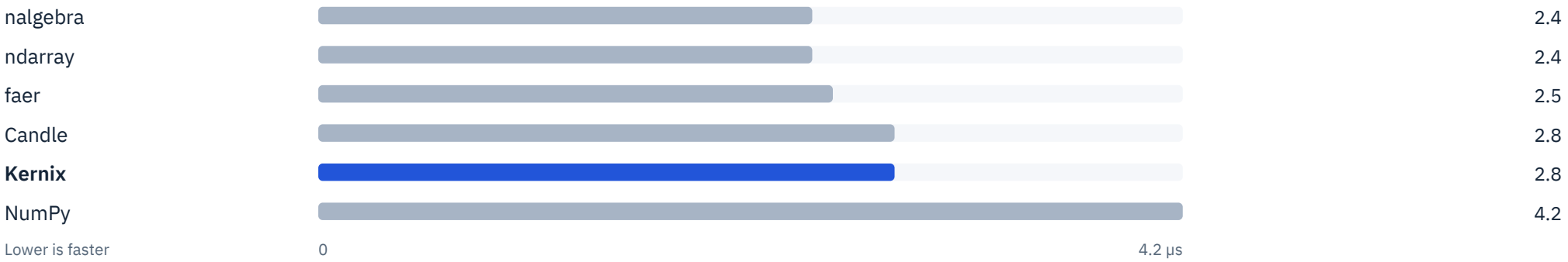
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

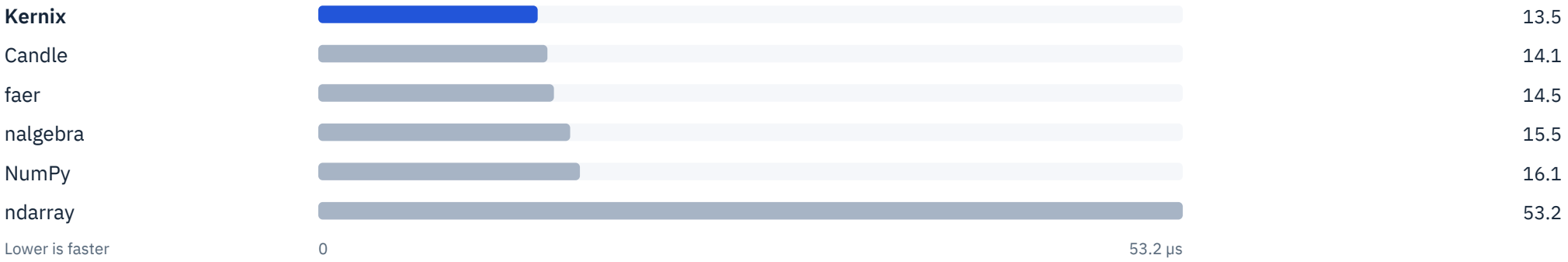
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



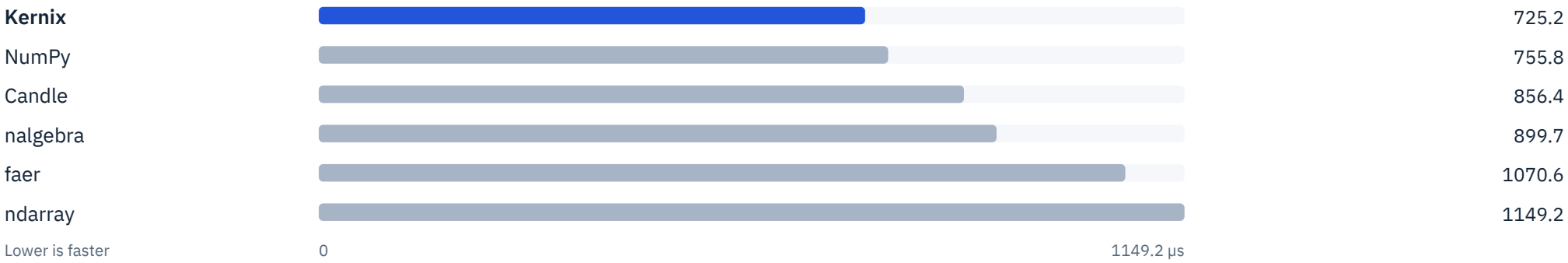
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



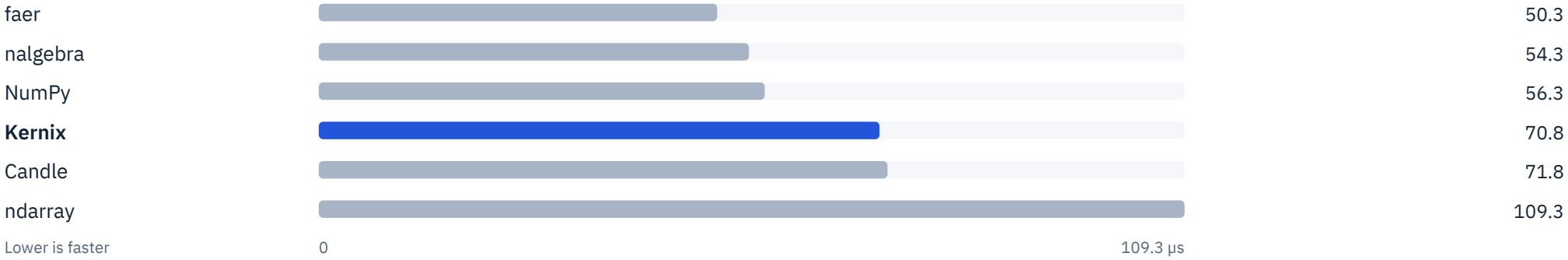
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

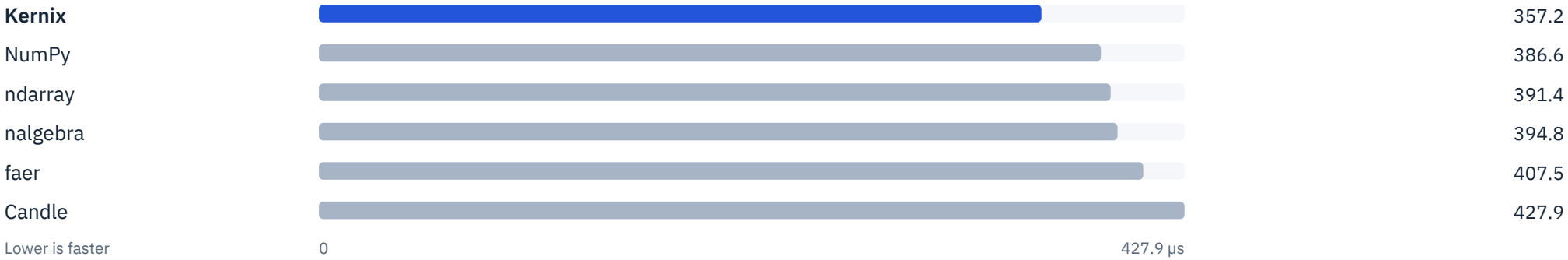
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



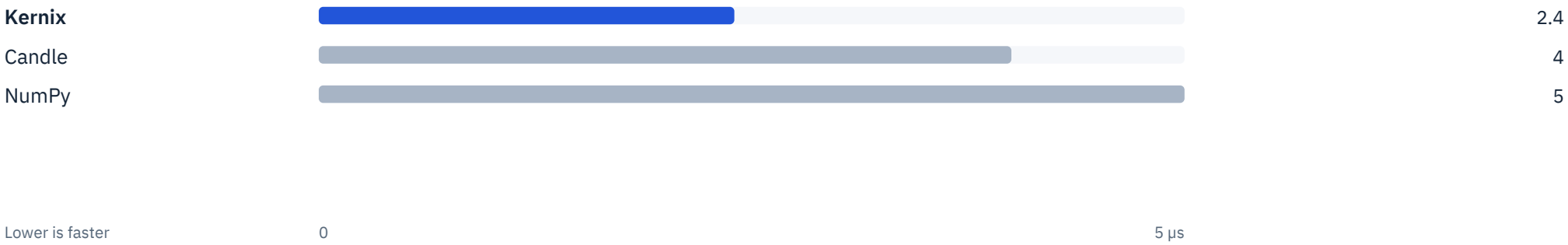
Kernix 0.11.1 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



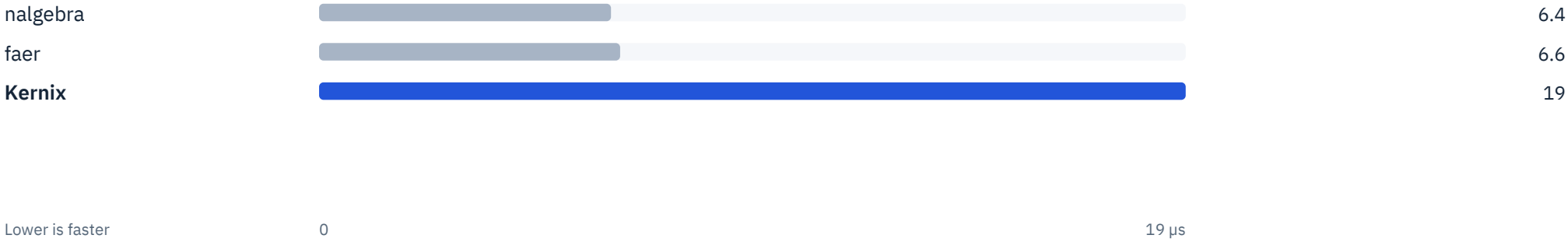
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

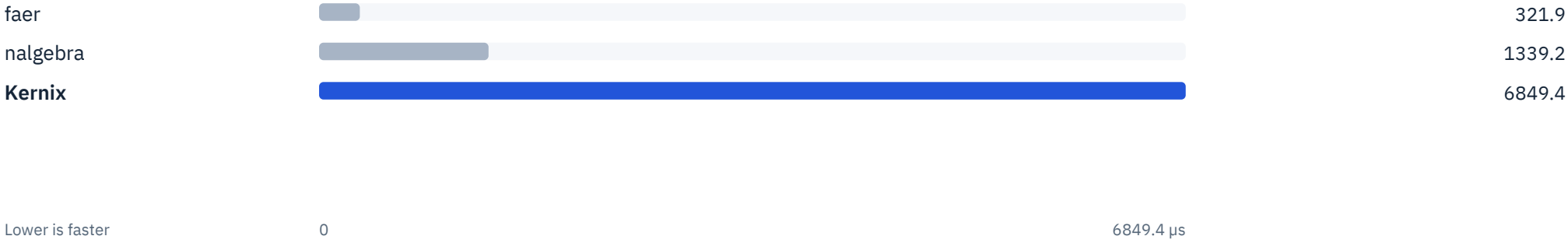
128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



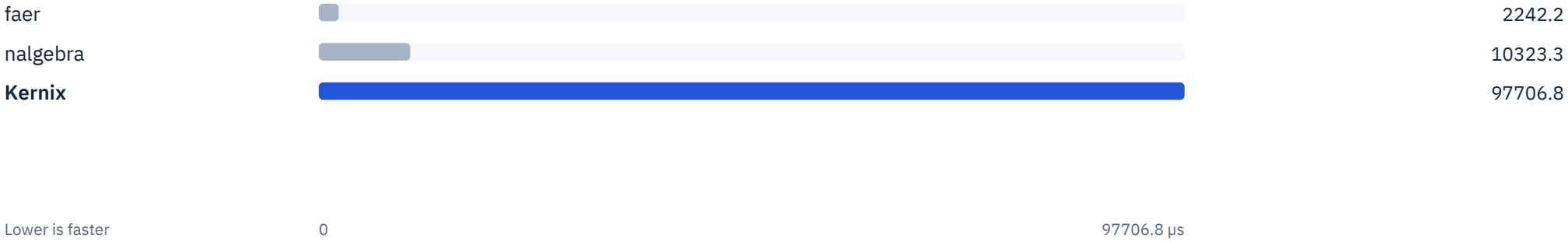
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

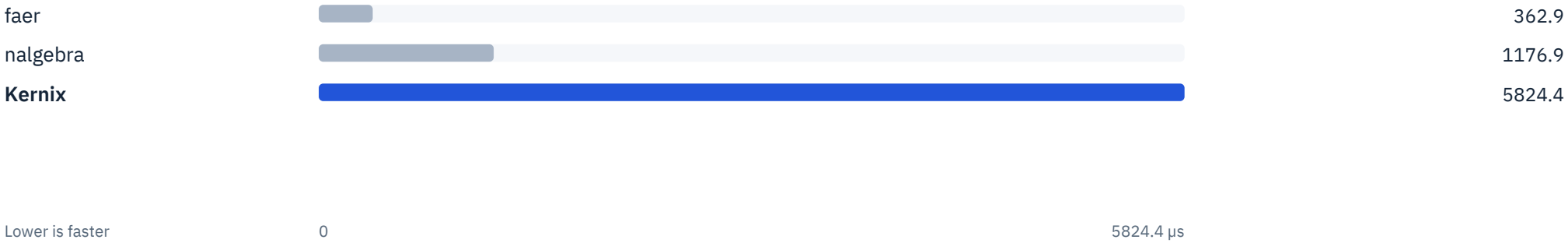
512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



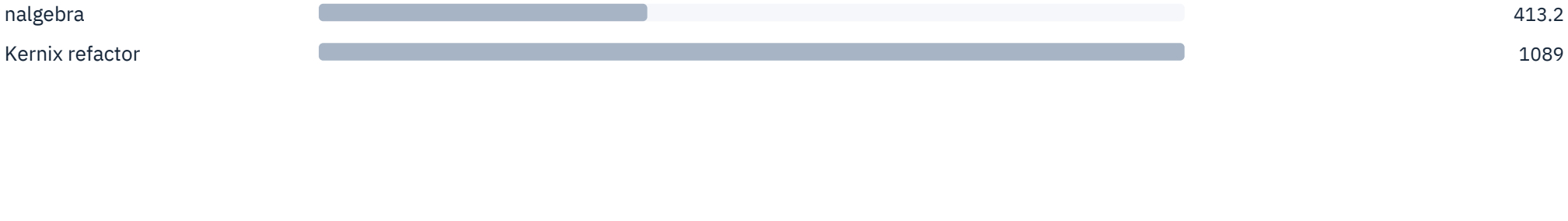
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

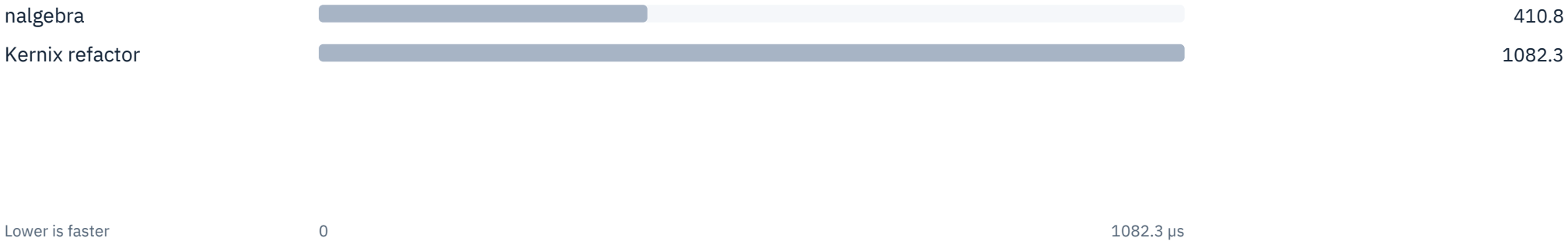
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



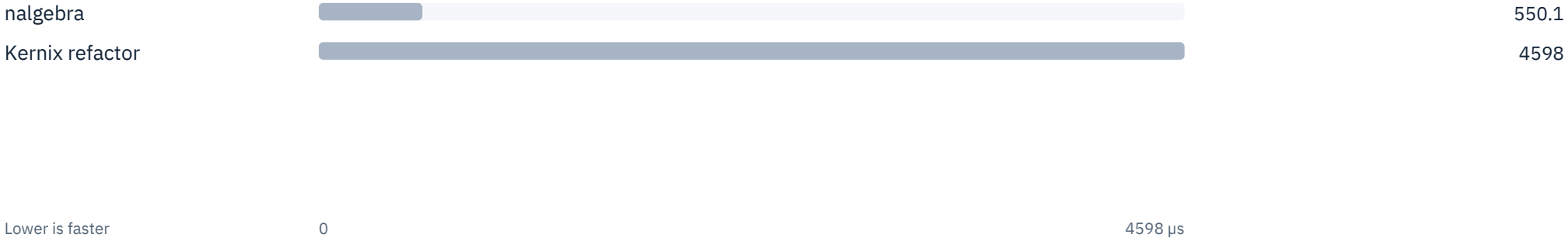
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median

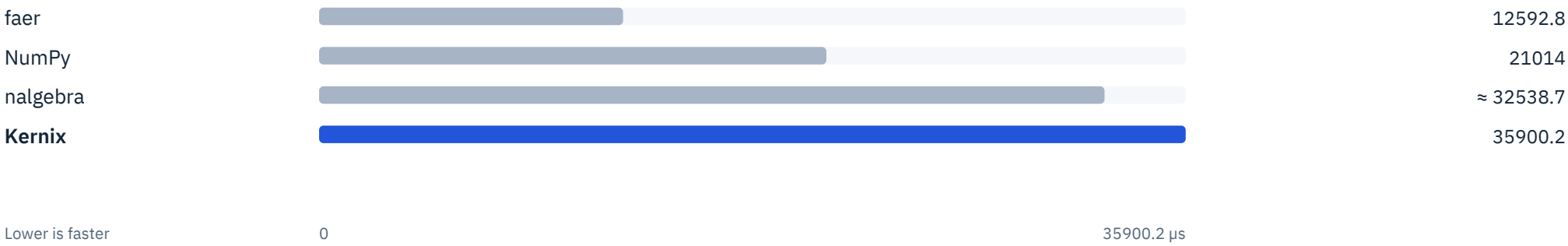


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

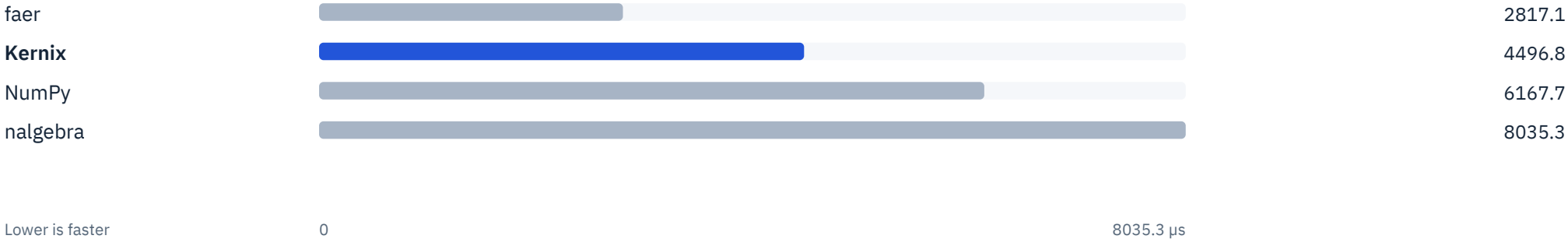
Thin QR

1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

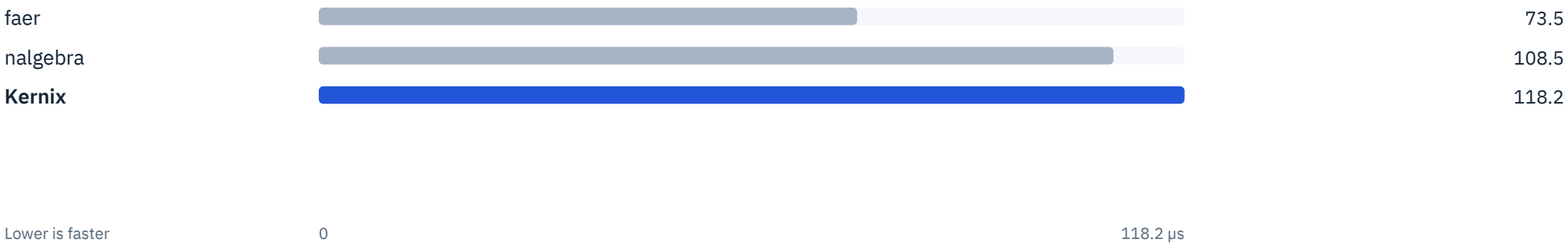


Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

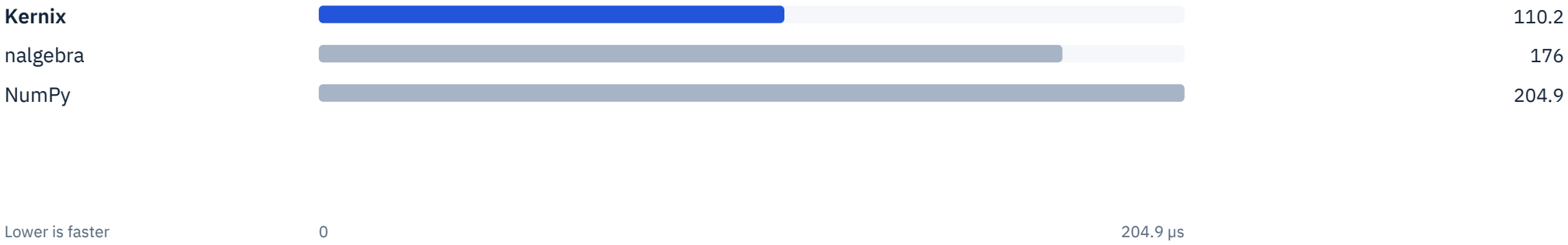
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

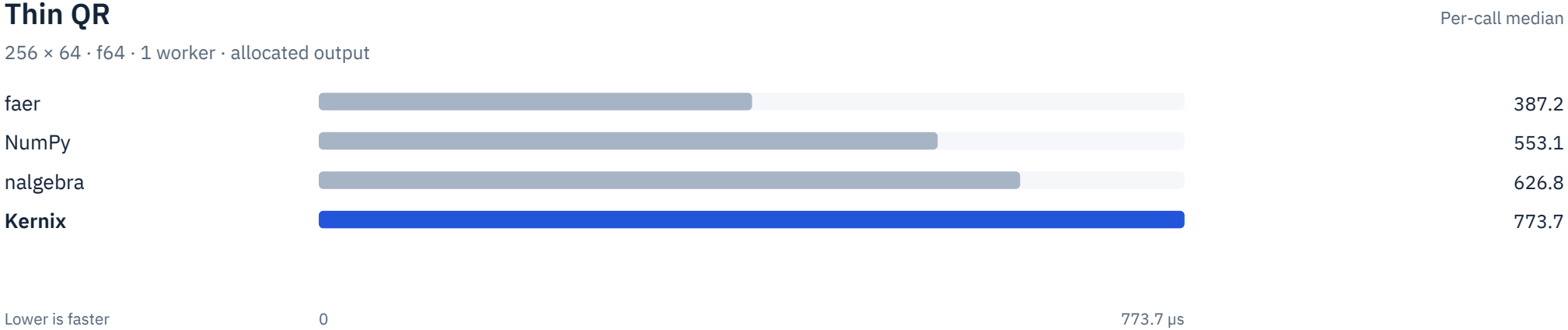
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



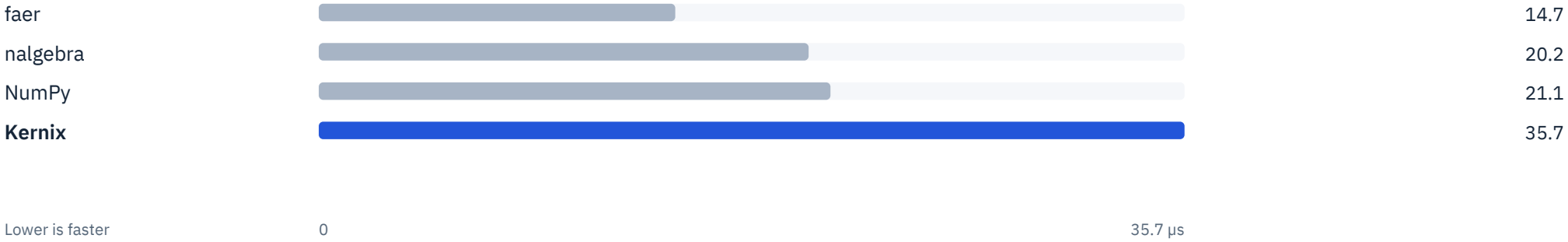
Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

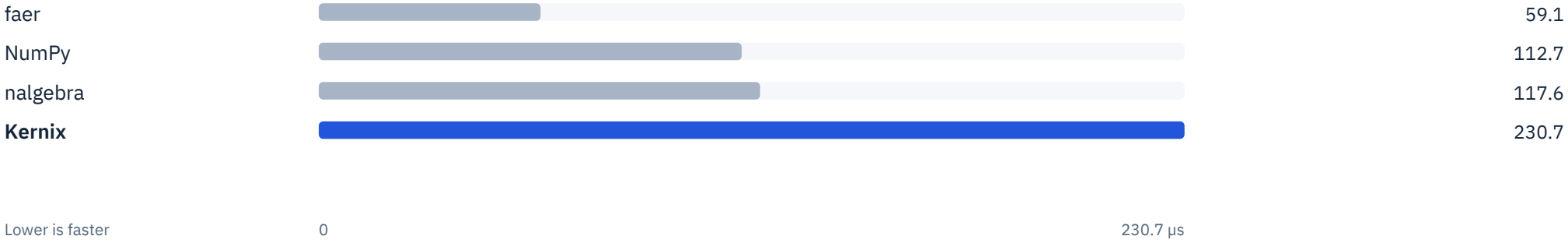
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



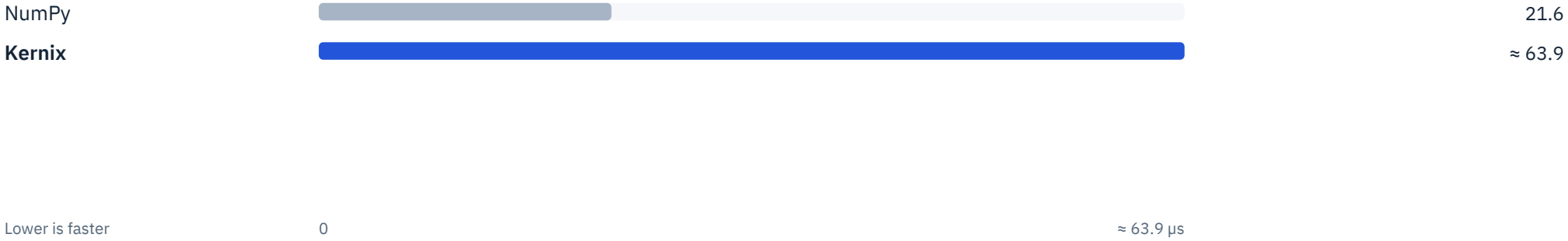
Kernix 0.11.1 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median

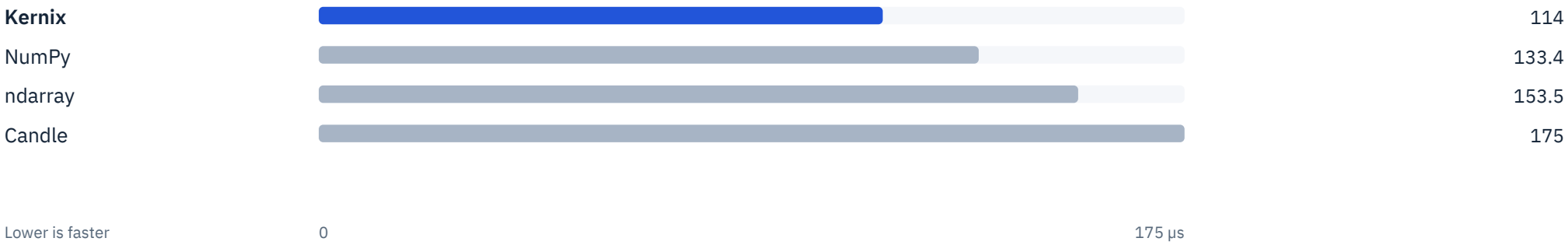


Kernix 0.11.1 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



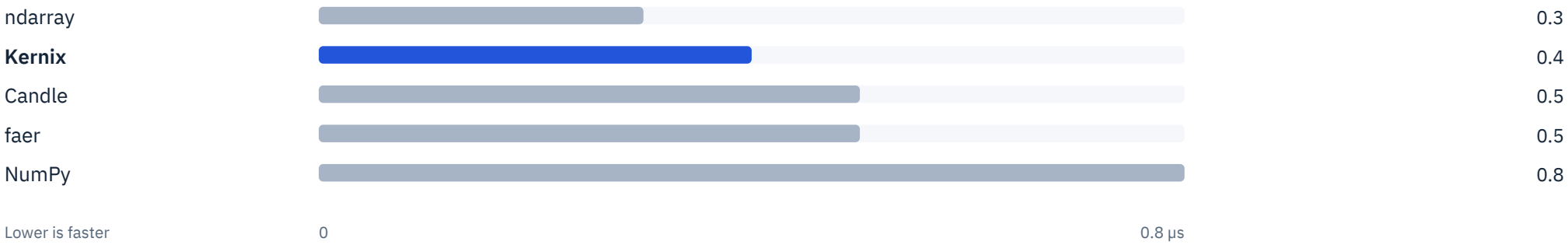
Kernix 0.11.1 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	107200	107.2
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	235200	235.2
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	60600	60.6
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	117900	117.9
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	138200	138.2
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	101800	101.8
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	75500	75.5
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56400	56.4
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2800	2.8
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4200	4.2
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	16100	16.1
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	53200	53.2
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14100	14.1
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15500	15.5
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13500	13.5
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14500	14.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	135400	135.4
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100300	100.3
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115800	115.8
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	136100	136.1
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	111100	111.1
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	114400	114.4
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	755800	755.8
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	899700	899.7
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	725200	725.2
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	856400	856.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1149200	1149.2
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1070600	1070.6
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7700.000000000001	7.700000000000001
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	27900	27.9
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10500	10.5
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	27000	27
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56300	56.3
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	109300	109.3

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	54300	54.3
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	71800	71.8
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	70800	70.8
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50300	50.3
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	394800	394.8
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	357200	357.2
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	407500	407.5
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	386600	386.6
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	427900	427.9
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	391400	391.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2400	2.4
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4000	4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	4800	4.8
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1400	1.4
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	4100	4.1
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	81500	81.5
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	83500	83.5
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	419800	419.8
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	57600	57.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	6600	6.6
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	6400	6.4
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	19000	19
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	6849400	6849.4
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1339200	1339.2
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	321900	321.9
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	97706800	97706.8
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2242200	2242.2
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	10323300	10323.3
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1176900	1176.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	5824400	5824.4
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	362900	362.9
Cholesky rank-one update / nalgebra 256 × 256 · f64	413200	413.2
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	1089000	1089
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	1082300	1082.3
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	410800	410.8
Cholesky rank-one update / nalgebra 512 × 512 · f64	550100	550.1
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4598000	4598
Thin QR / nalgebra 512 × 128 · f64	4301900	4301.9
Thin QR / NumPy 512 × 128 · f64	4273200	4273.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 512 × 128 · f64	2098300	2098.3
Thin QR / Kernix 512 × 128 · f64	4861300	4861.3
Thin QR / Kernix 1024 × 256 · f64	35900200	35900.2
Thin QR / faer 1024 × 256 · f64	12592800	12592.8
Thin QR / NumPy 1024 × 256 · f64	21014000	21014
Thin QR / nalgebra 1024 × 256 · f64	32538699.999999996	32538.699999999996
Cholesky factorization / nalgebra 512 × 512 · f64	8035300	8035.3
Cholesky factorization / faer 512 × 512 · f64	2817100	2817.1
Cholesky factorization / Kernix 512 × 512 · f64	4496800	4496.8
Cholesky factorization / NumPy 512 × 512 · f64	6167700	6167.7

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 96 × 32 · f64	108500	108.5
Column-pivoted QR / faer 96 × 32 · f64	73500	73.5
Column-pivoted QR / Kernix 96 × 32 · f64	118200	118.2
Column-pivoted QR / Kernix 256 × 64 · f64	903100	903.1
Column-pivoted QR / faer 256 × 64 · f64	395100	395.1
Column-pivoted QR / nalgebra 256 × 64 · f64	905400	905.4
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	97200	97.2
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103800	103.8
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	113500	113.5
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	176000	176

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	204900	204.9
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	110200	110.2
Thin QR / faer 96 × 32 · f64	87100	87.1
Thin QR / NumPy 96 × 32 · f64	76600	76.6
Thin QR / nalgebra 96 × 32 · f64	78300	78.3
Thin QR / Kernix 96 × 32 · f64	98300	98.3
Thin QR / nalgebra 256 × 64 · f64	626800	626.8
Thin QR / Kernix 256 × 64 · f64	773700	773.7
Thin QR / faer 256 × 64 · f64	387200	387.2
Thin QR / NumPy 256 × 64 · f64	553100	553.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 64 × 64 · f64	21100	21.1
Cholesky factorization / Kernix 64 × 64 · f64	35700	35.7
Cholesky factorization / nalgebra 64 × 64 · f64	20200	20.2
Cholesky factorization / faer 64 × 64 · f64	14700	14.7
Cholesky factorization / NumPy 128 × 128 · f64	112700	112.7
Cholesky factorization / faer 128 × 128 · f64	59100	59.1
Cholesky factorization / Kernix 128 × 128 · f64	230700	230.7
Cholesky factorization / nalgebra 128 × 128 · f64	117600	117.6
Cholesky factorization / nalgebra 256 × 256 · f64	853400	853.4
Cholesky factorization / faer 256 × 256 · f64	313400	313.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 256 × 256 · f64	918200	918.2
Cholesky factorization / Kernix 256 × 256 · f64	701300	701.3
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	49200	49.2
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10200	10.2
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	9900	9.9
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4500	4.5
Model inference / Kernix batched normalized/sliced MLP · f64	3500	3.5
Model inference / Candle batched normalized/sliced MLP · f64	8100	8.1
Model recording + forward / Kernix batched normalized/sliced MLP · f64	5000	5
Model recording + forward / Candle batched normalized/sliced MLP · f64	10300	10.3

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	6100	6.1
Model backward / Candle prepared graph → fresh gradients · f64	32100	32.1
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	21600	21.6
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63899.99999999999	63.89999999999999
Row softmax / ndarray 256 × 64 · f64	153500	153.5
Row softmax / Candle 256 × 64 · f64	175000	175
Row softmax / Kernix 256 × 64 · f64	114000	114
Row softmax / NumPy 256 × 64 · f64	133400	133.4
Create zeros / Kernix 64 × 64 · f32	400	0.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	800	0.8
Create zeros / Candle 64 × 64 · f32	500	0.5
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / faer 64 × 64 · f32	500	0.5
Create ones / faer 64 × 64 · f32	500	0.5
Create ones / NumPy 64 × 64 · f32	2100	2.1
Create ones / Kernix 64 × 64 · f32	500	0.5
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / ndarray 64 × 64 · f32	400	0.4